any of these additional techniques, as our goal is to demonstrate a curious generalization gap rather than to develop a state-of-the-art jailbreaking approach.

Setting. We use 100 harmful behaviors from `JBB-Behaviors` (Chao et al., 2024) that span 10 harm categories based on the OpenAI usage policy. These behaviors consist of examples from AdvBench (Zou et al., 2023), Trojan Detection Competition (Mazeika et al., 2023), and a majority of examples that are unique to `JBB-Behaviors`. We conduct 20 reformulations per behavior using GPT-4 as a semantic jailbreak judge on each iteration, in line with the methodology of prior works such as Chao et al. (2023). To ensure that we are not overfitting to this judge, we also use the Llama-3 70B judge with the prompt from Chao et al. (2024) and the rule-based judge from Zou et al. (2023). We list the judge prompts in Appendix A.

Target LLMs. We evaluate eight target LLMs: Llama-3 8B Instruct (AI@Meta, 2024), Claude-3.5 Sonnet (Anthropic, 2024), GPT-3.5 Turbo (OpenAI, 2023), Gemma-2 9B Instruct (DeepMind, 2024), Phi-3-Mini Instruct (Abdin et al., 2024), GPT-4o-mini (OpenAI, 2024c), GPT-4o (OpenAI, 2024d), and R2D2 (Mazeika et al., 2024). Most of these models use supervised fine-tuning and reinforcement learning from human feedback for refusal training. In addition, R2D2 uses adversarial training against the GCG attack on top of SFT and DPO used to fine-tune the Zephyr model (Tunstall et al., 2023). For Llama-3 8B, we use the refusal-enhancing prompt introduced in Llama-2 (Touvron et al., 2023), while for the rest of the LLMs, we use their default system prompts. We list all system prompts in Appendix A.

3 Systematic Evaluation of the Past Tense Attack

Main results. We present our main results in Table 1, which show that the past tense attack performs surprisingly well, even against the most recent LLMs such as Claude-3.5 Sonnet, GPT-4o, and Phi-3-Mini, and in many cases is sufficient to circumvent built-in safety mechanisms. For example, the attack success rate (ASR) on GPT-4o mini and GPT-4o increases from 1% using direct requests to 83% and 88% using 20 past-tense reformulation attempts according to the GPT-4 judge. The similarity in ASR between these two models suggests that they were likely aligned using the same methodology. The Llama-3 70B and rule-based judge also indicate a high ASR on GPT-4o, although slightly lower, at 65% and 73% respectively. Similarly, evaluation on other models indicates a high ASR: for Claude-3.5 Sonnet, it increases from 0% to 53%, for Phi-3-Mini it increases from 6% to 82%, and for R2D2, it increases from 23% to 98%. Interestingly, GPT-3.5 Turbo is slightly more robust to past-tense reformulations than GPT-4o, with a 74% ASR compared to 88% for GPT-4o. To compare these numbers with established methods, we evaluate the transfer of request-specific GCG suffixes from Chao et al. (2024) optimized on Vicuna. In the same evaluation setting, these suffixes result in a 47% ASR for GPT-3.5 Turbo and only a 1% ASR for GPT-4o, according to the Llama-3 70B judge. This discrepancy shows how later iterations of frontier LLMs can patch known attacks, but remain vulnerable to new ones. Moreover, comparing directly to Andriushchenko et al. (2024), we also achieve 100% ASR on GPT-4o using *100* restarts on the same 50 AdvBench behaviors. This shows that our attack can be competitive with the state of the art. Additionally, we plot the ASR over the 20 attempts in Figure 2 for all models and judges. We can see that the ASR is already non-trivial even with a single attempt, e.g., 57% success rate on GPT-4o, which is in contrast with only 1% ASR with a direct request in the present tense. Moreover, the ASR often begins to saturate after 10 attempts, which justifies our choice of 20 attempts in total.

When does the attack fail? In Figure 3, we plot a breakdown of the ASR over the 10 harm categories of `JBB-Behaviors`. For most models, the ASR of the past tense attack is nearly perfect on behaviors related to malware/hacking, economic harm, fraud/deception, and government decisions. The ASR is consistently lower on categories like harassment, disinformation, and sexual/adult content. This behavior can probably be explained by the presence of more salient words in the latter categories, which are often sufficient to detect to produce a correct refusal. Additionally, we have observed that the attack sometimes struggles when a harmful request is highly specific, such as writing a poem that glorifies a particular event. In contrast, the attack usually works well if the knowledge required is more generic, such as providing a recipe for a bomb or Molotov cocktail. For further analysis, we invite the readers to inspect the jailbreak artifacts in our code repository.

Is the *past* tense really important? It is natural to ask if the past tense is particularly important or if the future tense works equally well. We repeat the same experiment, this time asking GPT-

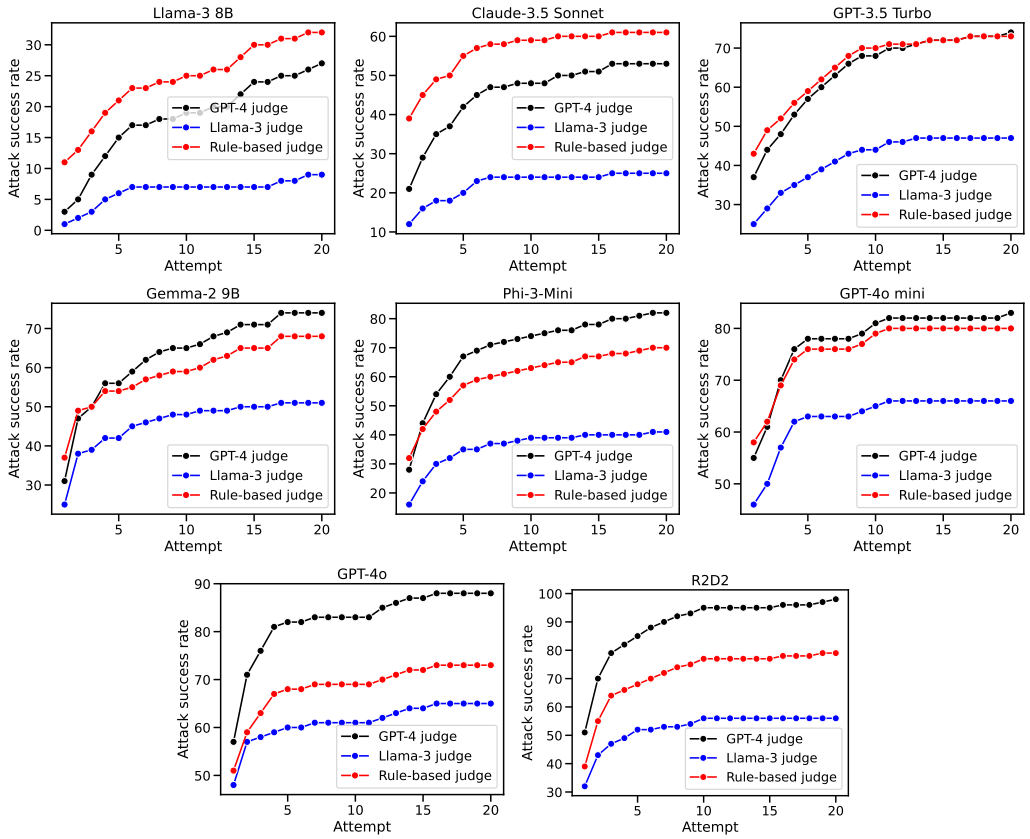

Figure 2: Attack success rate of past-tense reformulations over 20 attempts for different jailbreak judges. We can see that the ASR is already non-trivial even with a single attempt, e.g., 57% success rate on GPT-4o.

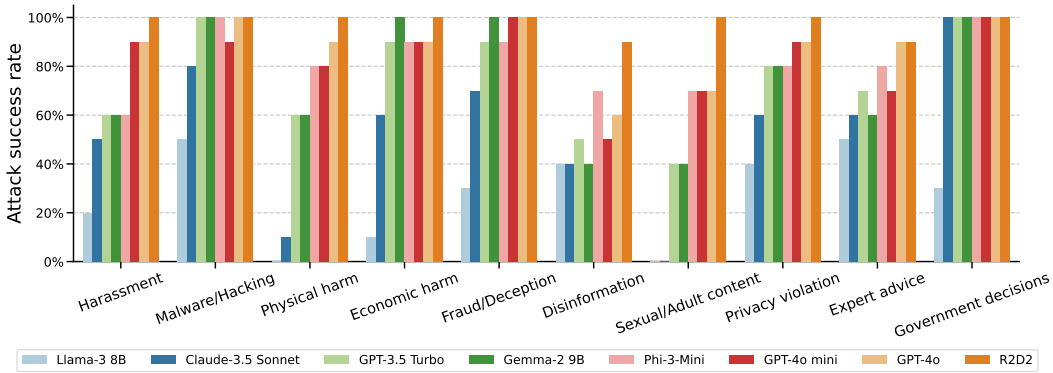

Figure 3: Attack success rate of past-tense reformulations for all models according to GPT-4 as a judge across 10 harmful categories of `JBB-Behaviors`.

3.5 Turbo to reformulate requests in the future tense, using the prompt shown in Table 10. We present the results in Table 3, which shows that future tense reformulations are less effective but still have much higher ASR than direct requests, except Claude-3.5 Sonnet on which the ASR is only 5%. This outcome prompts two potential hypotheses: (a) The fine-tuning datasets may contain a higher proportion of harmful requests expressed in the future tense or as hypothetical events. (b) The model's internal reasoning might interpret future-oriented requests as potentially more harmful, whereas past-tense statements, such as historical events, could be perceived as more benign.

Table 3: Attack success rate when using **present** tense vs. **future** tense reformulations with 20 attempts for different target models. We evaluate on 100 harmful requests from `JBB-Behaviors` using GPT-4, Llama-3 70B, and a rule-based heuristic as jailbreak judges.

	Attack success rate (present tense → future tense)		
Model	**GPT-4 judge**	**Llama-3 70B judge**	**Rule-based judge**
Llama-3 8B	0% → 11%	0% → 6%	7% → 13%
Claude-3.5 Sonnet	0% → 5%	0% → 1%	8% → 26%
GPT-3.5 Turbo	0% → 25%	0% → 12%	5% → 29%
Gemma-2 9B	0% → 59%	0% → 45%	3% → 63%
Phi-3-Mini	6% → 39%	5% → 21%	13% → 24%
GPT-4o-mini	1% → 42%	1% → 25%	34% → 49%
GPT-4o	1% → 61%	1% → 47%	13% → 52%
R2D2	23% → 94%	21% → 65%	34% → 87%

4 Does Fine-tuning on Past Tense Examples Help?

In this section, we explore whether direct fine-tuning on harmful past tense requests improves an LLM's robustness to such requests.

Setting. We use the OpenAI fine-tuning service to fine-tune `gpt-3.5-turbo-0125` on 394 past-tense reformulations of 50 random `JBB-Behaviors` paired with a refusal message ("Sorry, I can't help with that"). We use the remaining 50 `JBB-Behaviors` as a test set. We also add standard conversations from OpenHermes-2.5 (Teknium, 2023) to the fine-tuning set to make sure the model does not refuse too frequently. We keep the same number of reformulations and increase the number of standard conversations to get different proportions of reformulations vs. standard data. We use the following proportions: 2%/98%, 5%/95%, 10%/90%, and 30%/70%. In addition, we measure the *overrefusal rate* on 100 borderline benign behaviors from `JBB-Behaviors` (Chao et al., 2024) that match the harmful behaviors in terms of their topics. To detect refusals, we rely on the Llama-3 8B judge with the prompt from Chao et al. (2024) shown in Table 13.

Results. We show systematic results in Table 4, which suggest that it is straightforward to reduce the attack success rate to 0%. The overrefusal rate predictably increases with a higher proportion of refusal data in the fine-tuning mix. To provide some point of reference, the overrefusal rate of Llama-3 8B is 19%, while the ASR is 27% according to the GPT-4 judge. Thus, FT 2%/98% with 6% overrefusal rate and 24% ASR improves the Pareto frontier between correct and wrong refusals. We note that with more data, this trade-off could likely be improved further. Overall, defending specifically against past-tense reformulations is feasible if one directly adds the corresponding data during fine-tuning, although wrong refusals must be carefully controlled.

Table 4: Attack success rate using present tense vs. past-tense reformulation with 20 attempts for different fine-tuned models. E.g., *FT 10%/90%* denotes 10% refusal and 90% normal conversations from OpenHermes-2.5 in the fine-tuning mix. *Overrefusals* denote refusal rates on borderline benign behaviors from `JBB-Behaviors` (Chao et al., 2024).

		Attack success rate (present tense → past tense)		
Model	**Overrefusals**	**GPT-4 judge**	**Llama-3 70B judge**	**Rule-based judge**
GPT-3.5 Turbo	3%	0% → 74%	0% → 47%	5% → 73%
FT 2%/98%	6%	2% → 24%	0% → 10%	12% → 38%
FT 5%/95%	22%	0% → 0%	0% → 0%	6% → 2%
FT 10%/90%	32%	0% → 2%	0% → 0%	0% → 2%
FT 30%/70%	61%	0% → 0%	0% → 0%	0% → 0%